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ORIGINAL RESEARCH

Estimation of the Current Sheath Dynamics and Magnetic Field for Theta Pinch by Snow Plow Model Simulation

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Abstract
The snowplow model and shock wave equation are used to simulate theta pinch plasma experiment. The model calculates the sheath velocity, radial position, plasma current, plasma temperature, and the induced magnetic field of the current sheath. The model uses two phases: the radial inward phase and the reflected shock phase. The governing equations are derived by using the current factor and the mass factor. The model calculates the induced magnetic field of the plasma by the deriving equation and the magnetic field of the coil. Also, the model is operated by using the Hydrogen, Deuterium and Tritium gases at the same of the mass density. The results illustrate that the temperature and the induced magnetic field are higher for tritium gas, which has a higher molecular weight, while the current sheath velocity of tritium is lower. Some results of the model were compared with Lee model and other paper.

Keywords Theta pinch · Current sheath · Shock wave · Induced magnetic field

Introduction

The induced plasma current is generated in a theta direction when a condenser bank discharges into a coil around a cylindrical tube, where its direction becomes opposite of the primary discharge current. The temperature and spectrum are 150 eV and 300 to 2000Å for VUV respectively at the energy of 1 kJ at least [1]. The density and temperature of the theta pinch are estimated by using a series of a triple probe, which equal to 3.0×10^{21} m^{-3} and 40 eV at 20 kV [2]. The pulsed discharge current of the coil is 18 kA and the energy transfer efficiency of 85% are obtained by the spherical theta pinch [3].

The electromagnetic force is used to generate, accelerate and confine the current sheath of plasma. Snowplow and slug models were used and developed to study the current sheath motion inside the axial and the radial phases for plasma focus by Lee [4, 5]. The shock wave properties are used to obtain the shock velocity and the plasma temperature [6], while the plasma circuit equation is used to obtain the discharge current and voltage across the tube [7]. Also the model is developed by adding the mass and current factors [8], and the reflected shock phase and the radiative phase to simulate the X-ray emission [9], and using a beam–target mechanism to obtain the neutron yield [10], it becomes able to estimate the lifetime of the pinch [11].

The model used a two coupling coefficients, the first factor (f_c) is the ratio between the induced plasma current and the primary current of the coil which equals to 0.2 in argon gas and the other factor (f_m) which is the mass swept up for theta pinch [12]. The momentum equation of plasma is derived and solved numerically by using Mathematica program. The code studied the effect of the (β) parameter and the discharge current on a normalized radius. The theta pinching effect is improved by increasing the discharge current and decreasing the (β) factor [13]. The EDDY code is used to calculate the inductance and magnetic field with discharge current in the theta direction in order to design the FRX-L device [14].

The two different factors are used. The mass factors (f_m) is the fraction of mass swept by the magnetic piston. The mass factor is used to reduce the difference between the calculated velocities and the actual measured experimentally. The mass factor depends on the current value and the

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initial neutral gas density [15]. The fraction of sweeping plasma mass to total plasma mass is named mass sweeping efficiency factor (f_m). Also, a lot of electrical current transmit from magnetic piston and the remaining flows from inner and outer surface. The current factor (f_c) is the fraction of current transmission in piston to current transmission of total plasma surface due to current shedding [16]. Both factors have different value according to the best fitting that is obtained for each experiment or in each input conditions like the gas type, initial gas pressure, input energy, and device dimensions.

The present work aims to simulate theta pinch experiment by using snowplow model and shock wave equation, where the model suggests that, there is a slug of plasma is formed between the current sheath and shock wave. Also, the magnetic field relation is deduced to obtain the induced magnetic field of the plasma.

Theta Pinch Device Description

The theta pinch is obtained by discharging a capacitor bank through the wide wrapped conducting loop around a cylindrical ceramic tube which is filled with a gas at low pressure. The pulsed primary current (I_c) generates a highly variable pulsed magnetic field and hence generates a potential difference which ionizes the gas producing an induced current loop on the inner surface of ceramic tube (I_p) in an opposite direction of the primary circuit current (I_c) which causes heating of a gas and producing a plasma as shown in Fig. 1 [1, 17]. The induced current interacts with the primary current to generate the magnetic force which pushes the current sheath (CS) radially towards the axis (away from the ceramic tube inner wall) acts as a magnetic piston. The electromagnetic force is generated as a result react between the two opposing current loops, it moves the current sheath loop radially towards the axis in order to compress it, this process is known as a radial phase. The motion is highly supersonic which generates a radial shock front (SF) moves ahead of the piston. The SF reaches the axis before the (CS) and hits it, then it reflects outwards, while the (CS) still moves towards the axis. The SF meets the (CS) at a pinching time point, this phase is called reflected phase. Figure 1 shows that, the primary loop current (I_c) produces a magnetic field pointing into inwards of the page in the area enclosed by the primary loop, while the induced current I_p (piston) produces a magnetic field pointing outwards of the page in the area enclosed by the induced current loop [12].

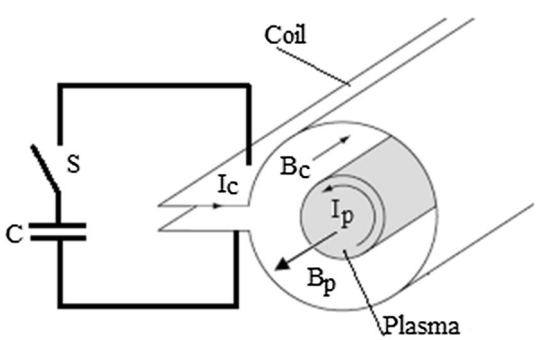

Fig. 1 Schematic diagram of theta pinch illustrates the primary current (I_c) of the coil and the induced plasma current (I_p)

Equations of the Model

Radial Inward Phase

Circuit Equation

The condenser bank discharges in the coils of theta pinch to generate the current I_c in theta direction which generates the induced magnetic flux in z-direction. The magnetic flux in a tube with (l_c) length and r_c radius is given by $\varphi_c = \mu\pi r_c^2 I_c / l_c$, where the magnetic field intensity in solenoid coil is given by $B_c = \mu I_c / l_c$. The axial magnetic flux (φ_c) generates the plasma current loop I_p which generates a reversed magnetic flux $\varphi_p = -\mu\pi r_p^2 I_p / l_c$. So, the net flux in the tube is given by:

$$\varphi_T = \varphi_c - \varphi_p = \frac{\mu\pi}{l_c}\left(r_c^2 I_c - r_p^2 I_p\right)$$

where r_p is the radius of the CS, so the inductance of theta pinch plasma can be estimated by:

$$L_p = \frac{\varphi_T}{I_c} = \frac{\mu\pi}{l_c}\left(r_c^2 - r_p^2 f_c\right) \tag{1}$$

where $f_c = I_p/I_c$ is the fraction of the current flowing in the piston due to the shedding effect. This factor is constant for theta pinch, where it depends on the fitting between the measured and calculated current waveforms [6, 12]. The general circuit equation can be written by using Kirchhoff's Law according to Fig. 2:

$$V_0 - \int \frac{I_c dt}{C_0} = \frac{dI_c}{dt}(L_0 + Lf_c) + I_c f_c \frac{dL}{dt} + r_0 I_c$$

$$V_0 - \int \frac{I_c dt}{C_0} = \frac{dI_c}{dt}\left(L_0 + f_c \frac{\mu\pi}{l_c}\left(r_c^2 - r_p^2 f_c\right)\right) - 2I_c f_c \frac{\mu\pi}{l_c} r_p \frac{dr_p}{dt} + r_0 I_c$$

$$\frac{dI_c}{dt} = V_0 - \int \frac{I_c dt}{C_0} + 2I_c f_c \frac{\mu\pi}{l_c} r_p \frac{dr_p}{dt} - r_0 I_c \Big/ \left(L_0 + f_c \frac{\mu\pi}{l_c}\left(r_c^2 - r_p^2 f_c\right)\right) \tag{2}$$

The circuit equation is constant for three phases, where C_o, L_o and r_o are the capacitance of capacitor, the circuit inductance and the stray resistance respectively.

Equation of the Motion

The momentum equation of theta pinch equals to the total force of the magnetic field and shock wave, which is given by:

$$F_T = \frac{d(mv)}{dt} = \frac{d}{dt}\left[\pi\rho l_c f_m\left(r_c^2 - r_p^2\right)\frac{dr_p}{dt}\right]$$

where r_c, r_p and l_c are the radius of the tube, the radius of the current sheath and the length of the tube, 'dr_p/dt' is the CS velocity, f_m is the fraction of mass swept by the current sheath motion and 'ρ' is the initial gas density, so

$$F_T = \pi\rho l_c f_m\left[\left(r_c^2 - r_p^2\right)\frac{d^2r_p}{dt^2} - 2r_p\left(\frac{dr_p}{dt}\right)^2\right] \quad (3)$$

The mass factor (f_m) is used to reduce the difference between the small values of the calculated velocity and the actual measured experimentally, due to the fact that the swept mass of the theoretical snowplow model is greater than the actual one [6]. The total magnetic pressure can be calculated by $P_T = P_{mb} - P_{mf}$, where 'P_{mb}' is the magnetic pressure behind the CS while 'P_{mf}' is the magnetic pressure in front of the CS, then, finally

$$P_T = P_{mb} - P_{mF} = \frac{\mu f_c(2 - f_c)}{2}\left(\frac{I_c}{l_c}\right)^2$$

The final magnetic force can be estimated by the following relation:

$$F_m = \int_0^r P_m dA = \int_0^r \frac{\mu f_c(2 - f_c)}{2}\left(\frac{I_c}{l_c}\right)^2 2\pi l_c dr$$

$$F_m = \frac{\mu\pi f_c(2 - f_c)}{l_c} r_c I_c^2 \quad (4)$$

The model uses the shock front which is moved ahead of the CS, and then reaches the axis and reflects towards the CS, which still moves towards the axis. The generated shock pressure compresses the CS to a minimum- non zero radius [18]. The force due to the shock wave may be given by:

$$F_s = A_s P_s = \frac{4\pi l_c \rho}{\gamma + 1} r_p v_s^2 \quad (5)$$

where $P_s = 2\rho v_s^2/(\gamma + 1)$ is the shock wave pressure, v_s is the shock wave velocity [19] and γ is the specific heat ratio for ideal gas [20]. From Eqs. 3, 4, and 5, the equation of motion can be written as:

$$\frac{d^2r_p}{dt^2} = \alpha^2 r_c I_c^2 - \xi r_p v_s^2 + 2\psi r_p\left(\frac{dr_p}{dt}\right)^2 \Big/ \psi\left(r_c^2 - r_p^2\right) \quad (6)$$

where $\alpha^2 = \frac{\mu\pi f_c(2-f_c)}{l_c}$, $\xi = \frac{4\pi d\rho f_m}{\gamma+1}$ and $\psi = \pi\rho l_c f_m$

Shock Wave Velocity and Plasma Temperature

According to the thermal dynamics, the shock wave velocity which moves ahead of the current sheath to sweep the neutral particles can be estimated by [21, 22]:

$$v_s = \left(\frac{\gamma + 1}{2}\right) v_p \quad (7)$$

From the shock wave relation, the plasma temperature (T) is given by [23, 24]:

$$T = \left(\frac{2M_W(\gamma - 1)}{R_0\, D(\gamma + 1)^2}\right) v_s^2 \quad (8)$$

where M is the molecule weight, R_o is the universal gas constant, D is the departure coefficient = D_N $(1 + Z_{eff})$; where $D_N = 2$ is the dissociation number for H, D, and T while $D_N = 1$ for Ar and He; Z_{eff} is the effective charge state [22, 23].

Magnetic Field Relation

The magnetic field is generated in z-direction, when the condenser bank discharges in the coil. This field induces the plasma current which generates an induced magnetic field in opposite direction. We suggested that the magnetic flux (Φ_c) (without plasma) across the area (A) with the magnetic field intensity (B_c) as shown in Fig. 3a is given by:

$$\Phi_c = B_c \cdot A_c = B_c \cdot \pi r_c^2$$

Also, in the presence a neutral gas, the magnetic flux (Φ_c) forms the plasma current which generates a reversed magnetic flux (Φ_p), where the plasma is diamagnetic. The magnetic flux (Φ_p) with plasma can be determined by the following relation:

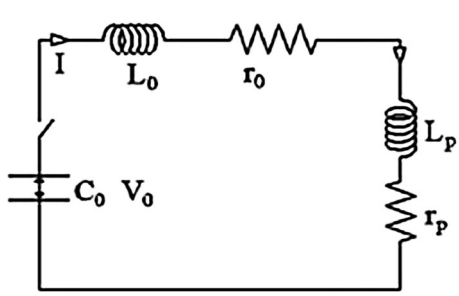

Fig. 2 The equivalent electric circuit of theta pinch

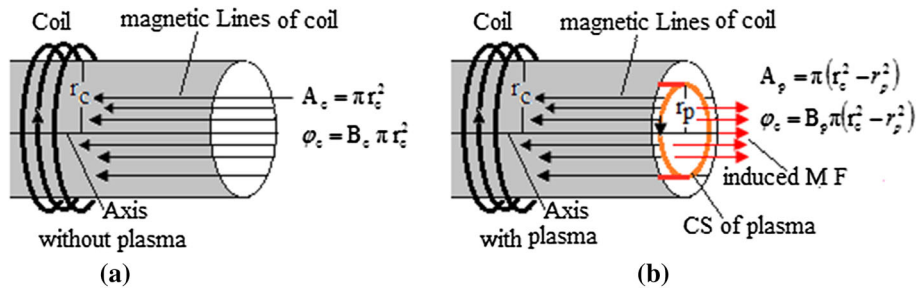

Fig. 3 Magnetic field distribution inside the tube **a** without plasma and **b** with plasma

$$\varphi_p = B_p \cdot \pi\,(r_c^2 - r_p^2)$$

where (B_p) is the induced magnetic field intensity of the current sheath. The model is assumed that the total magnetic flux is constant with and without plasma, so

$$-\varphi_p + \varphi_c = 0$$

$$\varphi_p = \varphi_c$$

$$B_p \cdot \pi(r_c^2 - r_p^2) = B_c \cdot \pi r_c^2$$

$$B_p = f_c B_c \cdot \frac{r_c^2}{(r_c^2 - f_m r_p^2)},\ r_p < r_c, \tag{9}$$

The magnetic field intensity can be calculated by previous relation, where f_m, f_c are used in the relation to express the real effective of both mass density which is swept in the radial direction, and effective magnetic field of CS respectively, where the general form of the relation has the same original shape.

Radial Reflected Shock Phase

The radial phase is finished when the SF hits the axis '$r_s = 0$', and reflects towards the CS, then the radial reflected phase starts. The shock front velocity in this phase is 0.3 times the shock front velocity in the end of radial phase. The radial reflected phase uses the same equations of radial phase except the shock front velocity [25]. The reflected shock velocity equals to:

$$v_{rs} = -0.3\,(v_s)_{on-axis} \tag{10}$$

where $(v_s)_{on-axis}$ is the velocity of the shock front when the SF hits the axis at the end of the radial phase and v_{rs} is the reflected shock speed [8]. In this phase, the current sheath is moved towards the axis by accompanied a virtual forward shock front moving with the on-axis speed of the incident shock. The reflected shock moves outwards with a fraction of the on-axis incident shock speed [19, 26, 27].

Results and Discussion

The code is operated by three phases, radial inward phase, radial reflected shock phase and expansion phase. Firstly, the code is operated at the same conditions of Lee code [12], where L_o = 338nH, C_o = 60 μF, r_o = 13 mΩ, r_c = 5.5 cm, l_c = 30 cm, V_0 = 10 kV, P_0 = 0.08 torr of argon gas, but in order to obtain the same data of Lee, the f_m and f_c are being 0.13, 0.2 respectively. Figure 4a illustrates a comparison between the current flows through the coil (I_c) and the induced plasma current (I_p) as a function of the discharge time with Lee model. The computed current of the coil and induced plasma current are agreement with the Lee published paper [12] which also are agreeing with published paper [17]. Also, Fig. 4b illustrates the induced plasma current (I_p) for different charging voltages. The figure shows that the plasma column is compressed, where the dips of the current is decreased with increasing the charging voltages. The pinch disappears with low voltage where the magnetic pressure is lower.

Figure 5 shows that the radius of CS and SF has a higher value at beginning of radial phase (radius of the tube), then it is decreased radially as a function of time in the radial phase. There is a slug formed between the CS and SF, where the SF moves quickly to sweep up the neutral particles ahead of CS, so the slug thickness increases due to a higher velocity of SF [22]. The reflected shock phase (RSF) starts, when the SF hits the axis at 5.3 μs then it reflects outwards, while the CS continue moves towards the axis. The slug thickness is decreased because the CS still moves towards the RSF until meets at 5.6 μs, then the plasma column is pinched by the magnetic and shock pressures [24]. The computed data agree with the Lee published paper [12]. In this model, the line and Bremsstrahlung radiation loss and Joule heating gain are negligible in this study.

Figure 6 illustrates the CS and SF velocities in the radial phase. It is clear that the SF velocity is bigger than the CS velocity, where the computed CS velocity 2.51 cm/μs is agreement with measured velocity in Chaisombata published paper [17]. This result is obtained at 12 kV, 0.02 torr of Ar gas and f_m = 0.8, f_c = 0.6.

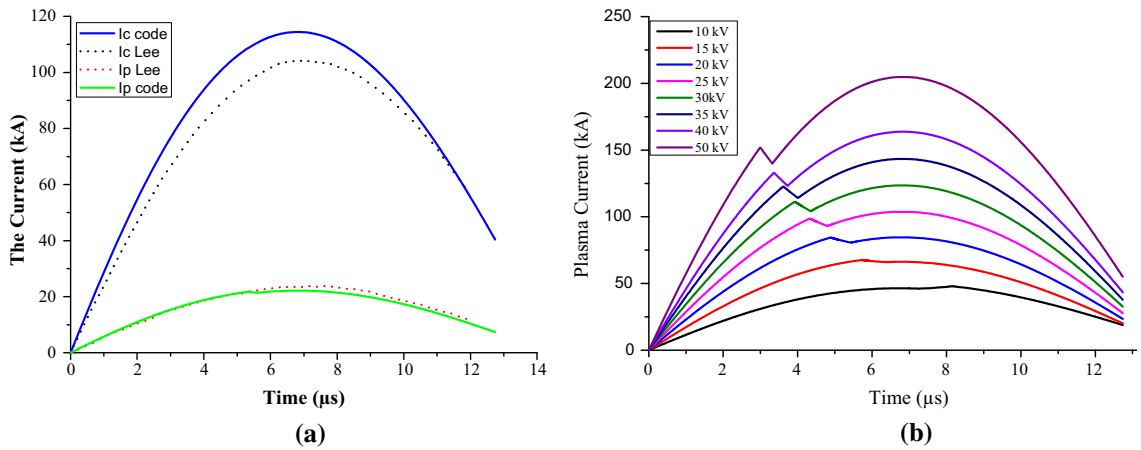

Fig. 4 **a** The current of both the coil I_c and induced plasma I_p for Lee code and our code, **b** the plasma current I_p for different voltage and $f_m = 0.6$, $f_c = 0.4$, P = 0.6 torr of Ar

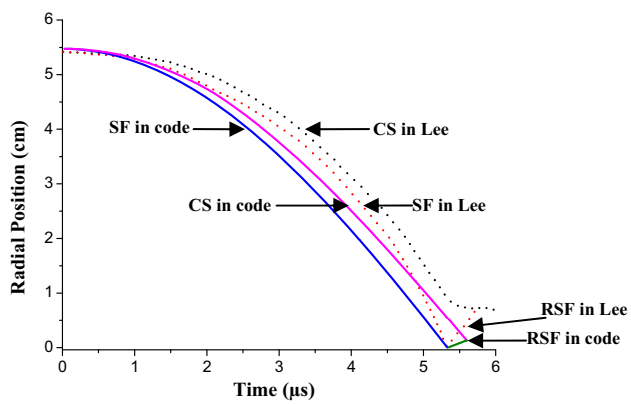

Fig. 5 The radial position of CS and SF for our code and Lee code

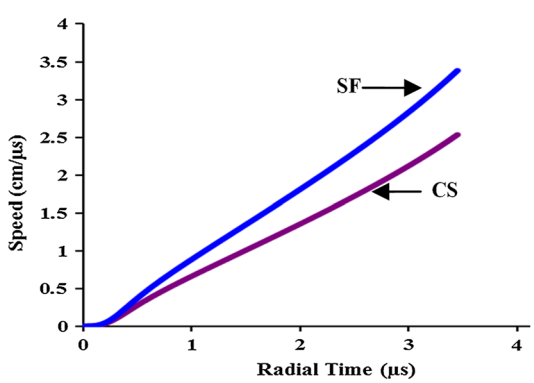

Fig. 6 The radial velocity of the CS and SF

Figure 7 illustrates the plasma temperature in the radial and reflected phases. The temperature is increased as a function of time towards the axis. At the same of previous conditions, the CS temperature is 4.7 eV, but in order to obtain the same data ($T_e = 3.91$ eV) which is agreement with the Chaisombata published paper [17], the current factor $f_c = 0.66$. The temperature is increased when the SF

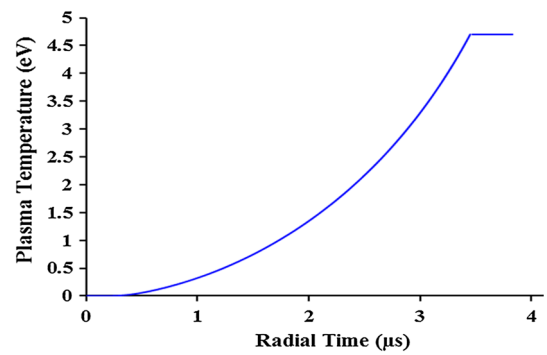

Fig. 7 The temperature of CS in the radial and reflected shock phases

velocity is increased until the SF hits the axis, then the reflected SF phase starts which is assumed that the RSF velocity is constant (see Eq. 10), so the temperature has a plateau as shown in Fig. 7. The temperature trajectory is similar the shock velocity, where the SF ionizes the gas particles ahead of the plasma slug, so it increases the plasma temperature.

Figure 8 illustrates the magnetic field intensity of the coil and plasma inside the discharge tube. The magnetic field intensity of coil (B_c) is calculated by using Ampere's law ($B_c = \mu I_c / l_c$), the calculated data agree with the current of coil (I_c) (see Fig. 4). Also, the figure shows the induced magnetic field of plasma B_p according to the derived relation (see Eq. 9), where the B_p is increased with the B_c, then it is decreased at the pinch time of 5 μs as shown in the waveform of current (see Fig. 4). A dip of the B_p shows that the current sheath is decreased because the magnetic pressure is bigger than the kinetic pressure. We noted that the B_p varies with the I_p variation (Ampere's law), but in this paper, the B_p does not depend on the I_p according to the deduced equation (relation 9).

Figure 8b shows the induced magnetic field and the radial position as a function of the time. To clarify the

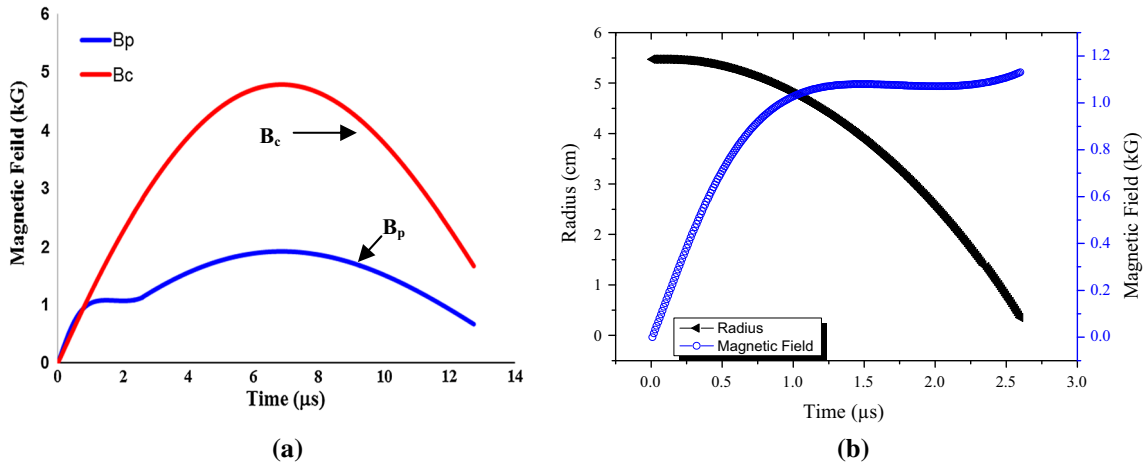

Fig. 8 **a** The magnetic field shape of coil and plasma with time, and **b** the magnetic field distribution B_p with radial position at 10 kV, 0.08 torr, $f_m = 0.7$, $f_c = 0.4$

variation of the magnetic field with the radius, we divided this variation into two separate curves; both are function of the discharge time as shown in the Fig. 8b. As the time of the discharge increases, the radius of the plasma sheath decreases in the same time that the magnetic field increases. Combining both curves together, one can find that at the beginning of the discharge (left hand side of the Fig. 8b), a slight change of the inward motion lead to slight decrease of the sheath radius combined with increasing of the magnetic field which is the expected behavior due to the increase of the discharge current (which is the same as the right hand side of Fig. 8b). After about 2.5µs, the sheath reaches a stage of the higher velocity on the same time at which the discharge current reached its maximum and starts to decrease. Since the magnetic field is a function of the current and the radial distance, then it will decrease as the current decrease, but increase as the radius of the sheath decreases. So it will be approximately constant as the right hand side of the figure, which is the same as the left hand side of Fig. 8 b).

Figure 9 shows the induced magnetic field (B_p) distribution inside the discharge chamber with variation of the charging voltage. The induced field (B_p) is increased with the charging voltage while the pinching time is decreased. These results show that the derived equation (Eq. 9) in order to estimate the induced magnetic field is valid and acceptable. The pinch phenomena disappear in the induced field shape at lower charging voltage as 10 kV.

Figure 10 shows the current sheath velocity and the temperature as a function of charging voltage and pressure. The figure shows that the velocity and temperature are increased with the charging voltage, while decreased with the gas pressure, where the CS mass is increased with the gas pressure and according to this model, the plasma slug between the CS and RF is increased with increasing of the gas pressure, where the SF sweeps up the particles ahead the CS.

Theta Pinch Parameters for H, D, and T

The code is operated by using the Hydrogen (H), Deuterium (D) and Tritium (T) gases. The variations of the CS velocity, temperature and magnetic field as a function of time and radial position at 10 kV, 0.08 torr are calculated as illustrated in Figs. 11, 12, 13 and 14. Figure 11 on the left hand shows the CS velocity in radial phase, where the CS velocity of the H is greater than the T gas, where the CS mass and plasma slug are smaller than the T gas. Also, the right figure has the same behavior, where the velocity of the H is greater than the T velocity. The molecular weight of the T gas is greater than of the H gas. Figure 12 illustrates the radial position of the plasma for H, D and T gases. The shock front reaches the axis at 2.69, 3.22 and 3.58 µs, while the SF meets the CS at pinch times 2.9, 3.48 and 3.94 µs for the H, D and T gases respectively. As the molecular weight of the gas and the plasma slug are increased, also the pinching times are increased and the time which the SF hits the CS is increased.

Figure 13 illustrates the plasma temperature for H, D, and T gases. The temperature of tritium gas is a higher than the temperature of H and D gases, its molecular weight, which direct proportional with the temperature as shown in Eq. 8 is a highest, so it is predicted that the density is a lower. For tritium gas, the molecular weight is larger, so the initial density of T gas is lower than the H and D gases at the same initial density. The higher densities of gases have a lower temperature, where the collisions of particles lead to a thermalization process between the gas species. On the other hand, the temperature depends on the ratio

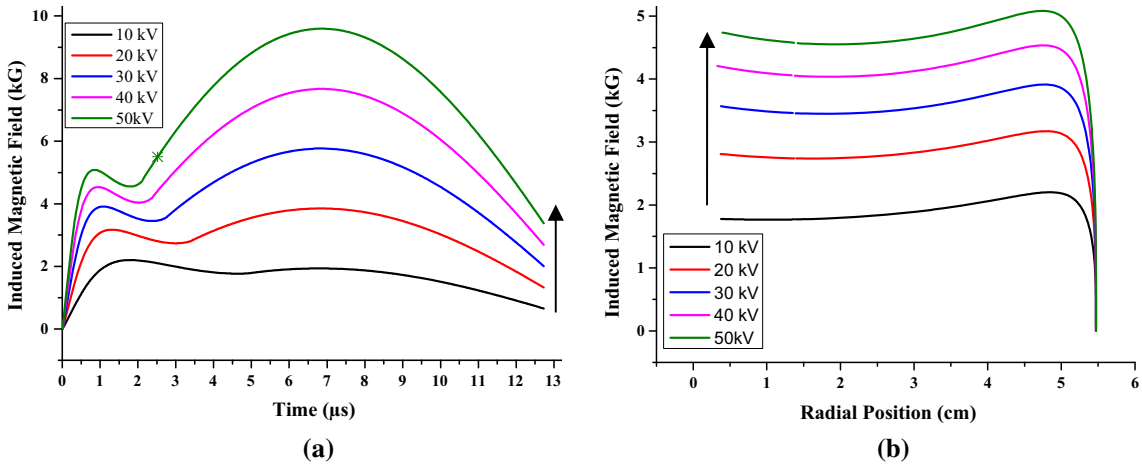

Fig. 9 **a** The induced magnetic field B_p at different voltages, and **b** the induced magnetic field distribution with the radial position at 0.08 torr of Ar, $f_m = 0.8$, $f_c = 0.4$

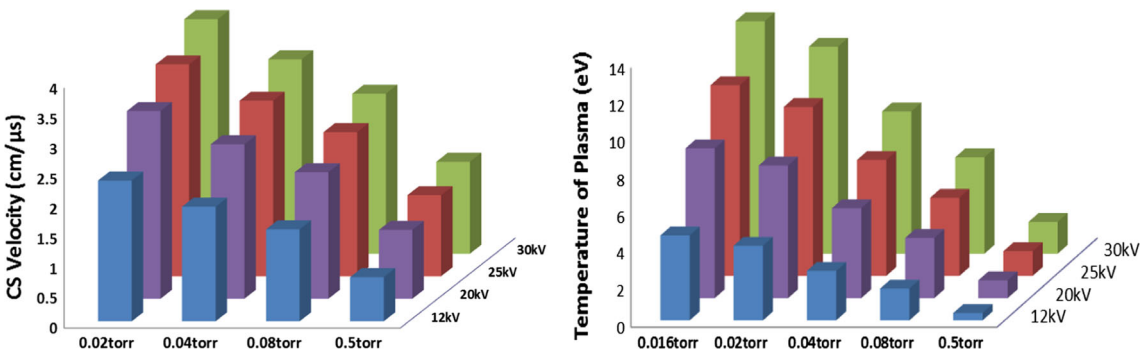

Fig. 10 The CS velocity and temperature as function of the gas pressure and charging voltage

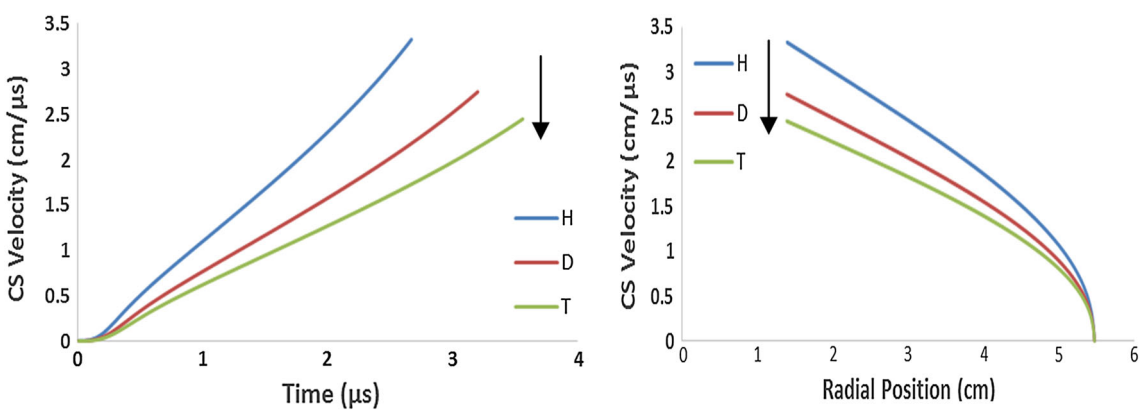

Fig. 11 The CS velocity for H, D and T gases at 10 kV and 0.08 torr

$(\gamma - 1)\big/(\gamma + 1)^2$ of the specific heat [22–28] as shown in Eq. 8. The effect of specific heat ratio is very smaller than the effect of molecular weight of gases, which equals to 1.4 for H, D, T gases. The figure shows that the temperatures are about 3.13 eV for T, 2.63 eV for D and 1.92 eV for H gases. Also, the effect of charge state is constant for H, D, and T. The right figure explains the distribution of the temperature inside the tube, where it is increased towards the axis.

Figure 14 illustrates the induced magnetic field (MF) for H, D and T gases at 10 kV and 0.08 torr. The graph shows that the radial and reflected shock phases only are used. The left figure shows that the field is a higher for T than the field for H and D gases. The temperature of T gas which

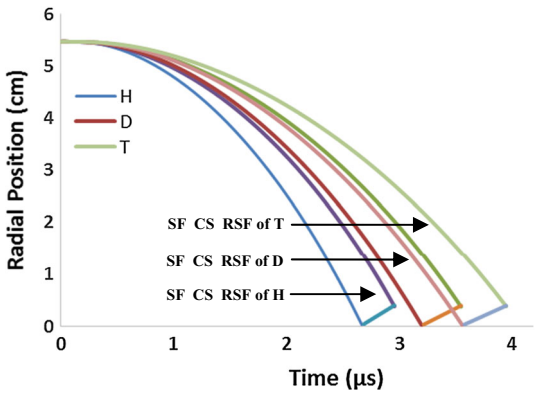

Fig. 12 The radial positions of H, D and T gases at 10 kV, 0.08 torr

depends on the molecular weight is a higher, so according to the Bennett relation I_p = cons. NKT [29], the induced MF is generated by the plasma current which depends on the temperature. The right figure shows the induced MF distribution inside the tube, which is increased towards the axis.

Conclusion

The theta pinch model is used to simulate the CS dynamic and the induced magnetic field of the plasma in two phases: a radial phase and reflected shock phase. The plasma slug between the SF and the CS is used to obtain the CS temperature of the plasma. The temperature, velocity and magnetic field were estimated for H, D and T gases at the same initial mass density. The data were compared with the theoretical Lee model and experimental paper of Chaisombata, where there is a good agreement. The simulation showed that the velocity of the H is higher than the velocity of the D and T gases, where the molecular weight of the H is lower than the D and H. Also, the derived equation of the induced magnetic field showed that the B_p is higher for T gas than the D and H gases, where the B_p depends on the current which is a function of the temperature according to Bennett relation, where the temperature depends on the molecular weight and SF velocity (see Eq. 8).

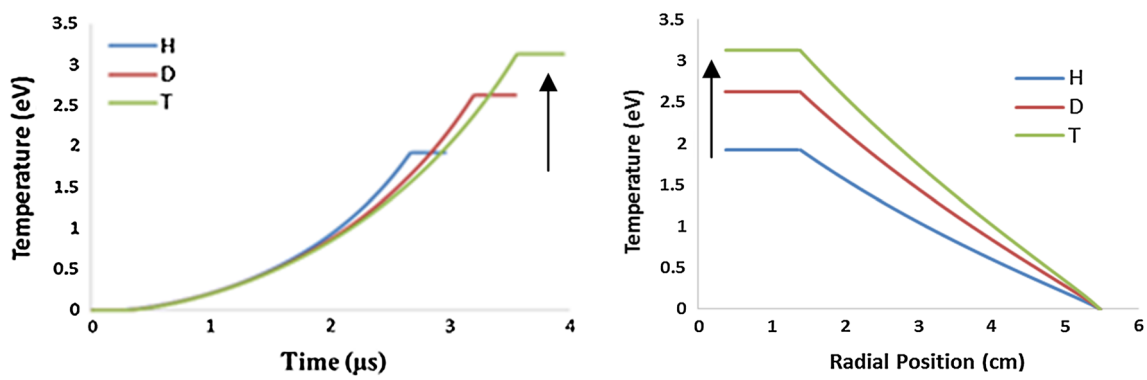

Fig. 13 The temperature of the H, D and T gases at 10 kV, 0.08 torr

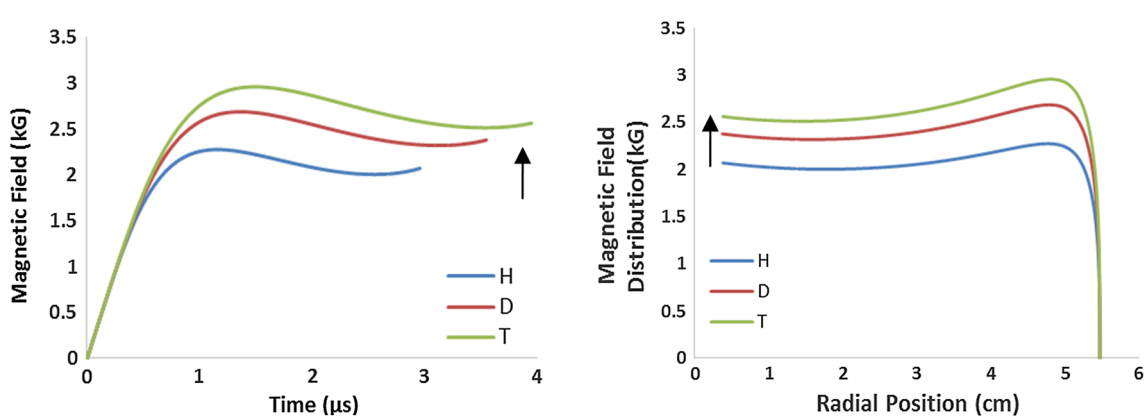

Fig. 14 The induced magnetic field of the H, D and T gases at 10 kV, 0.08 torr

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